# RANS PART 2: REYNOLDS-AVERAGED MOMENTUM EQUATION (X-COMPONENT)

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Fluid Dynamics and IBM Theory • Module 5

Adapted from a lecture by Steven L. Brunton.

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

# WHERE WE ARE (FROM PART 1)

We start from incompressible Navier–Stokes (nondimensional):

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}}\Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

Reynolds decomposition (time average):

$$\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}) + \boldsymbol{u}'(\boldsymbol{x}, t), \qquad \overline{\boldsymbol{u}'} = 0.$$

**Result from Part 1:**

$$\nabla \cdot \overline{\boldsymbol{u}} = 0, \qquad \nabla \cdot \boldsymbol{u}' = 0.$$

# OTHER IDENTITIES FROM PART 1

Recall we are taking averages in time. We know

$$\begin{aligned}
\overline{u+v} &= \overline{u} + \overline{v}, \\
\overline{\overline{u}} &= \overline{u}, \\
\overline{\overline{u} \cdot v} &= \overline{u} \cdot \overline{v}, \\
\overline{\frac{\partial u}{\partial x}} &= \frac{\partial \overline{u}}{\partial x}
\end{aligned}$$

In general,

$$\begin{aligned}
\overline{u \cdot v} &\neq \overline{u} \cdot \overline{v}, \\
\overline{u'^2} &\neq 0.
\end{aligned}$$

# MOTIVATION: WHY AVERAGE THE MOMENTUM EQUATION?

Engineers often want **mean** quantities:

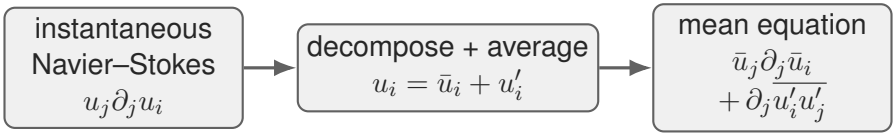
- mean drag / mean lift,
- mean separation behavior,
- bulk transport trends,

without resolving every turbulent eddy in space-time.

**Plan today**

Plug $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$ (and $p = \overline{p} + p'$) into momentum, then time-average to get an equation for $\overline{\boldsymbol{u}}$.

# WHAT THE AVERAGING OPERATION CHANGES




Schematic: instructor's course notes. Index notation: $u_i$ ($i = 1, 2, 3$) are the components $(u, v, w)$, $\partial_j = \partial/\partial x_j$, and a repeated index is summed.

**The nonlinear term does not simply average away**

The correlation $R_{ij} = \overline{u_i' u_j'}$ is new information. An equation for the mean velocity alone does not determine it: this is the RANS closure problem.

# WE WILL DERIVE ONLY THE X-MOMENTUM EQUATION

Write $\boldsymbol{u} = (u, v, w)^T$. The x-component of momentum is:

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\Delta u.$$

We substitute:

$$u = \overline{u} + u', \quad v = \overline{v} + v', \quad w = \overline{w} + w', \quad p = \overline{p} + p'.$$

# AVERAGING IDENTITIES WE WILL USE REPEATEDLY

Time average operator $\overline{\cdot}$ is linear and (under mild regularity) commutes with spatial derivatives:

$$\overline{a+b} = \overline{a} + \overline{b}, \qquad \overline{\frac{\partial}{\partial x} a} = \frac{\partial}{\partial x} \overline{a}.$$

Key facts:

$$\overline{u'} = 0, \qquad \overline{p'} = 0, \qquad \overline{\Delta u'} = \Delta \overline{u'} = 0.$$

**Critical warning (the source of Reynolds stresses)**

In general,

$$\overline{ab} \neq \overline{a}\,\overline{b}.$$

So averages of *products of fluctuations* need not vanish.

# STEP 1: SUBSTITUTE DECOMPOSITION INTO X-MOMENTUM

Start with

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + w\,\frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\Delta u.$$

Substitute $u = \overline{u} + u'$, $v = \overline{v} + v'$, $w = \overline{w} + w'$, $p = \overline{p} + p'$. Then

$$\frac{\partial}{\partial t}(\overline{u} + u') + (\overline{u} + u')\frac{\partial}{\partial x}(\overline{u} + u') + (\overline{v} + v')\frac{\partial}{\partial y}(\overline{u} + u') + (\overline{w} + w')\frac{\partial}{\partial z}(\overline{u} + u') = -\frac{\partial}{\partial x}(\overline{p} + p') + \frac{1}{\mathrm{Re}}\Delta(\overline{u} + u').$$

# STEP 1A: EXPAND THE CONVECTION TERM (X-PART SHOWN)

The first convection piece expands like multiplying polynomials:

$$(\overline{u}+u')\frac{\partial}{\partial x}(\overline{u}+u') = \overline{u}\,\frac{\partial \overline{u}}{\partial x} + \overline{u}\,\frac{\partial u'}{\partial x} + u'\,\frac{\partial \overline{u}}{\partial x} + u'\,\frac{\partial u'}{\partial x}.$$

Similarly,

$$(\overline{v}+v')\frac{\partial}{\partial y}(\overline{u}+u') = \overline{v}\,\frac{\partial \overline{u}}{\partial y} + \overline{v}\,\frac{\partial u'}{\partial y} + v'\,\frac{\partial \overline{u}}{\partial y} + v'\,\frac{\partial u'}{\partial y},$$

and likewise for the $w$-term.

## STEP 2: TIME-AVERAGE THE ENTIRE EQUATION

Time-average term-by-term.

**Time derivative:**

$$\overline{\frac{\partial}{\partial t}(\overline{u}+u')} = \frac{\partial}{\partial t}\overline{u} + \frac{\partial}{\partial t}\overline{u'} = \frac{\partial}{\partial t}\overline{u}.$$

For a statistically stationary mean, $\frac{\partial}{\partial t}\overline{u} = 0$. Otherwise this mean-acceleration term must remain.

**Pressure and viscosity:**

$$\overline{-\frac{\partial}{\partial x}(\overline{p}+p')} = -\frac{\partial}{\partial x}\overline{p}, \qquad \overline{\Delta(\overline{u}+u')} = \Delta\overline{u}.$$

**Mixed mean–fluctuation products die:** e.g.

$$\overline{\overline{u}\,\frac{\partial u'}{\partial x}} = \overline{u}\,\overline{\frac{\partial u'}{\partial x}} = 0, \quad \overline{u'\,\frac{\partial \overline{u}}{\partial x}} = \overline{u'}\,\frac{\partial \overline{u}}{\partial x} = 0.$$

What survives are *mean–mean* terms and *fluctuation–fluctuation* terms.

# STEP 2 RESULT: AVERAGED X-MOMENTUM (STATIONARY MEAN)

Here $\partial_x = \partial/\partial x$ etc.; subscripts are reserved for components.
After averaging, the surviving terms are:

$$\overline{u}\,\partial_x\overline{u} + \overline{v}\,\partial_y\overline{u} + \overline{w}\,\partial_z\overline{u} \;+\; \overline{u'\,\partial_x u'} + \overline{v'\,\partial_y u'} + \overline{w'\,\partial_z u'} = -\partial_x\overline{p} + \frac{1}{\mathrm{Re}}\Delta\overline{u}.$$

**Interpretation**

Everything is now in terms of mean fields *except* the three correlation terms $\overline{u'\,\partial_x u'}$, $\overline{v'\,\partial_y u'}$, $\overline{w'\,\partial_z u'}$. These encode how fluctuations force the mean.

# THE PROBLEMATIC SURVIVORS: WHY THEY DON'T VANISH

Even though $\overline{u'} = 0$, products of fluctuations need not average to zero:

$$\overline{(u')^2} \neq 0, \qquad \overline{u'v'} \neq 0, \qquad \overline{u'w'} \neq 0.$$

These are (components of) the **Reynolds stresses**. They will become **spatial derivatives of** $\overline{(u')^2}$, $\overline{u'v'}$, $\overline{u'w'}$.
This is where closure enters: we need a model for these correlations.

# STEP 3: THE STREAMWISE PRODUCT-RULE IDENTITY

Use the product rule:

$$\partial_x\big((u')^2\big) = 2u'\,\partial_x u' \quad \Rightarrow \quad u'\,\partial_x u' = \frac{1}{2}\,\partial_x\big((u')^2\big).$$

Therefore,

$$\overline{u'\,\partial_x u'} = \frac{1}{2}\,\frac{\partial}{\partial x}\overline{(u')^2}.$$

This half-factor is correct for this term in isolation. In the sum of all three convective terms, continuity supplies the complementary half, as the next two slides show explicitly.

## STEP 3: PRODUCT-RULE SETUP FOR THE CROSS TERMS

Apply the product rule to $u'v'$ and $u'w'$:

$$\frac{\partial}{\partial y}(u'v') = v'\,\partial_y u' + u'\,\partial_y v', \qquad \frac{\partial}{\partial z}(u'w') = w'\,\partial_z u' + u'\,\partial_z w'.$$

Rearrange:

$$v'\,\partial_y u' = \frac{\partial}{\partial y}(u'v') - u'\,\partial_y v', \qquad w'\,\partial_z u' = \frac{\partial}{\partial z}(u'w') - u'\,\partial_z w'.$$

Now time-average:

$$\overline{v'\,\partial_y u'} = \frac{\partial}{\partial y}\overline{u'v'} - \overline{u'\,\partial_y v'}, \quad \overline{w'\,\partial_z u'} = \frac{\partial}{\partial z}\overline{u'w'} - \overline{u'\,\partial_z w'}.$$

These identities expose derivatives of the correlations, but they also leave two terms involving derivatives of $v'$ and $w'$. The complete vector product rule organizes all three directions at once.

# STEP 3: PACKAGE THE FULL CONVECTIVE SUM

Apply $a_j \partial_j b = \partial_j(a_j b) - b\,\partial_j a_j$ with $a_j = u'_j$ and $b = u'$:

$$u'\,\partial_x u' + v'\,\partial_y u' + w'\,\partial_z u' = \frac{\partial (u')^2}{\partial x} + \frac{\partial (u'v')}{\partial y} + \frac{\partial (u'w')}{\partial z} - u'\left(\partial_x u' + \partial_y v' + \partial_z w'\right).$$

**Why the half-factor disappears**

As on the streamwise slide: the product-rule correction supplies the second copy of $u'\,\partial_x u'$, so the first term is $\partial_x(u'^2)$, not $\frac{1}{2}\partial_x(u'^2)$.

In detail: $\partial_x(u'^2) = 2u'\partial_x u'$ contains one copy too many; the extra copy is cancelled by the $-u'\partial_x u'$ inside $-u'(\nabla \cdot \boldsymbol{u}')$. That whole bracket vanishes only because all three of its pieces are kept together, which is why the cross terms must be treated in the same step.

# THE CANCELLATION: USE $\nabla \cdot \boldsymbol{u}' = 0$

The correction in the packaged identity is

$$\overline{u'\,\partial_x u'} + \overline{u'\,\partial_y v'} + \overline{u'\,\partial_z w'}.$$

Factor out $u'$:

$$u'\,\partial_x u' + u'\,\partial_y v' + u'\,\partial_z w' = u'\,(\partial_x u' + \partial_y v' + \partial_z w') = u'\,(\nabla \cdot \boldsymbol{u}').$$

But from Part 1, $\nabla \cdot \boldsymbol{u}' = 0$, hence

$$\overline{u'\,\partial_x u'} + \overline{u'\,\partial_y v'} + \overline{u'\,\partial_z w'} = 0.$$

The correction vanishes, so no factor of $\frac{1}{2}$ is missing (see the streamwise slide).

# STEP 3 RESULT: THE REYNOLDS-STRESS-GRADIENT FORM

Using the identities and cancellations, the fluctuation terms become:

$$\overline{u'\,\partial_x u'} + \overline{v'\,\partial_y u'} + \overline{w'\,\partial_z u'} = \frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}.$$

Equivalently, in index notation,

$$\overline{u_j'\partial_j u_i'} = \partial_j \overline{u_i' u_j'} - \overline{u_i'\partial_j u_j'} = \partial_j R_{ij},$$

because $\partial_j u_j' = 0$.

# FINAL X-COMPONENT RANS EQUATION (STATIONARY MEAN)

The Reynolds-averaged x-momentum equation can be written as:

$$\overline{u}\,\partial_x\overline{u} + \overline{v}\,\partial_y\overline{u} + \overline{w}\,\partial_z\overline{u} = -\partial_x\overline{p} + \frac{1}{\mathrm{Re}}\Delta\overline{u} - \Big(\frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}\Big).$$

**Key interpretation**

The mean flow is driven by pressure + viscosity *and* the divergence of Reynolds stresses.

# THE GENERAL UNSTEADY RANS EQUATION

If the averaging operation produces a time-dependent mean, retain its time derivative:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{1}{\mathrm{Re}} \Delta \bar{u}_i - \frac{\partial R_{ij}}{\partial x_j}, \qquad \frac{\partial \bar{u}_j}{\partial x_j} = 0.$$

**Stationary specialization**

The preceding derivation set $\partial_t \bar{u}_i = 0$ only after explicitly assuming statistically stationary mean fields. Reynolds averaging itself does not require this term to vanish.

# VECTOR/TENSOR FORM (WHAT THIS LOOKS LIKE IN COMPACT NOTATION)

Define the velocity-correlation tensor:

$$\boldsymbol{R} := \overline{\boldsymbol{u}'\,\boldsymbol{u}'^{T}} = \begin{bmatrix} \overline{u'u'} & \overline{u'v'} & \overline{u'w'} \\ \overline{v'u'} & \overline{v'v'} & \overline{v'w'} \\ \overline{w'u'} & \overline{w'v'} & \overline{w'w'} \end{bmatrix}.$$

Then the stationary mean momentum equation is commonly written:

$$(\overline{\boldsymbol{u}} \cdot \nabla)\overline{\boldsymbol{u}} = -\nabla \overline{p} + \frac{1}{\mathrm{Re}} \Delta \overline{\boldsymbol{u}} - \nabla \cdot \boldsymbol{R}.$$

Along with mean incompressibility: $\nabla \cdot \overline{\boldsymbol{u}} = 0.$

# CORRELATION TENSOR VERSUS TURBULENT STRESS

The term $-\nabla \cdot \boldsymbol{R}$ acts like an additional forcing in the nondimensional mean equation, with the same differential form as a stress divergence.

- $R_{ij} = \overline{u_i' u_j'}$ is a velocity-correlation tensor.
- In dimensional form, the corresponding turbulent (Reynolds) stress is commonly defined as $\tau_{ij}^R = -\rho R_{ij}$; some texts instead call $\rho R_{ij}$ the Reynolds-stress tensor and keep the minus sign in the equation.
- These are not determined by the mean equation alone.

**Closure problem (core of RANS modeling)**

The averaged equations introduce new unknowns ($\boldsymbol{R}$) that require modeling:

$$\boldsymbol{R} \approx \mathcal{M}(\overline{\boldsymbol{u}}, \nabla \overline{\boldsymbol{u}}, \ldots).$$

# WHAT COMES NEXT (CLOSURE + ML)

Next topics in a RANS sequence:

- classic closures (e.g., eddy viscosity / Boussinesq hypothesis),
- $k$–$\varepsilon$, $k$–$\omega$, Spalart–Allmaras, etc.
- data-driven closure ideas (ML approximations of $\boldsymbol{R}$ or corrections).

**Takeaway**

RANS reduces space-time complexity, but pays a price: **closure modeling**.

## WORKED CHECK: FULLY DEVELOPED CHANNEL FLOW

Take a statistically stationary, fully developed channel, $\overline{u} = \overline{u}(y)$, $\overline{v} = \overline{w} = 0$, with statistics independent of $x$ and $z$.

- The left side $\overline{u}\,\partial_x \overline{u} + \overline{v}\,\partial_y \overline{u} + \overline{w}\,\partial_z \overline{u}$ vanishes term by term.
- Of the Reynolds-stress terms only $\partial_y \overline{u'v'}$ survives, so the final x-equation becomes

$$0 = -\partial_x \overline{p} + \frac{1}{\mathrm{Re}} \frac{\mathrm{d}^2 \overline{u}}{\mathrm{d}y^2} - \frac{\mathrm{d}}{\mathrm{d}y} \overline{u'v'}.$$

- The y-equation gives $\overline{p} + \overline{v'^2}$ independent of $y$, so $\partial_x \overline{p}$ is a constant and the *total* shear stress $\frac{1}{\mathrm{Re}} \frac{\mathrm{d}\overline{u}}{\mathrm{d}y} - \overline{u'v'}$ is linear in $y$.

**Interpretation**

Near the wall the viscous part carries the stress; away from it the Reynolds shear stress $-\overline{u'v'}$ does. This is the starting point of the wall-scaling and log-law slides in deck 5-3 (see also Pope 2000, §7.1).

# SUMMARY

- Start with x-momentum component of incompressible Navier–Stokes.
- Substitute $\boldsymbol{u} = \overline{\boldsymbol{u}} + \boldsymbol{u}'$ and time-average.
- Mixed mean–fluctuation terms vanish; products of fluctuations remain.
- Use product-rule identities + $\nabla \cdot \boldsymbol{u}' = 0$ to package surviving terms as

$$-\left(\frac{\partial}{\partial x}\overline{(u')^2} + \frac{\partial}{\partial y}\overline{u'v'} + \frac{\partial}{\partial z}\overline{u'w'}\right) = -(\nabla \cdot \boldsymbol{R})_x.$$

- The resulting mean equation requires a closure model for $\boldsymbol{R}$.

# REFERENCES AND SIGN CONVENTION


- O. Reynolds, “On the dynamical theory of incompressible viscous fluids and the determination of the criterion,” *Philosophical Transactions of the Royal Society A*, 186 (1895). Pages 123–164; doi:10.1098/rsta.1895.0004.
- S. B. Pope, *Turbulent Flows*, Cambridge University Press (2000), Chapters 3–4.
- S. L. Brunton, lecture on Reynolds-averaged Navier–Stokes equations. YouTube (the video linked from Part 1; the derivation sequence of this deck is adapted from Brunton’s RANS lecture material).
- D. C. Wilcox, *Turbulence Modeling for CFD*, 3rd ed., DCW Industries (2006), Ch. 2 (Reynolds averaging and the Reynolds-stress tensor; alternative sign conventions).


**Convention used here**

$R_{ij} = \overline{u_i' u_j'}$ is the velocity-correlation tensor, so the nondimensional RANS equation contains $-\partial_j R_{ij}$. In dimensional stress notation, $\tau_{ij}^R = -\rho R_{ij}$ and the same contribution is $\rho^{-1}\partial_j \tau_{ij}^R$.